Sensitivity Analysis

Ramp Count vs Blockage Events

1. Experiment Setup

A fixed random seed (random-seed 42) was added to the setup procedure. This ensures the obstacle layout is identical across all five runs, so ramp count is the only variable affecting results.

Parameter	Value used
obstacle-density	0.20
ramp-search-radius	8 patches
num-general	15 agents
num-wheelchair	8 agents
num-elderly	8 agents
Random seed	42 (fixed — identical obstacle layout every run)
Simulation duration	Runs until all agents finish or 2000 ticks

Variable across runs: ramp-count set to 0, 5, 10, 15, and 20 in sequence

2. Results

Values recorded directly from NetLogo monitors at the end of each run.

ramp-count	Total blockages	% blocked	Avg time — wheelchair	Avg time — general	Accessibility gap
0	**156**	**100%**	140.0 ticks	59.5 ticks	80.5 ticks
5	0	0%	132.5 ticks	59.5 ticks	73.0 ticks
10	0	0%	135.9 ticks	59.5 ticks	76.3 ticks
15	0	0%	128.0 ticks	59.5 ticks	68.5 ticks
20	0	0%	131.8 ticks	59.5 ticks	72.2 ticks

Accessibility gap = mean wheelchair travel time minus mean general pedestrian travel time. General pedestrian time is consistent at 59.5 ticks across all runs because the obstacle layout is fixed.

3. What the Results Show

Three-zone pattern

The results show a clear three-zone pattern:

Zone	ramp-count	What happens
Exclusion	0	156 blockages, 100% of wheelchair users blocked, many do not complete the corridor
Transition	5	Zero blockages, all wheelchair users complete the journey, accessibility gap 73.0 ticks
Saturation	10 – 20	No further improvement, travel time stays between 128.0 and 135.9 ticks

Finding 1 — Non-linear blockage threshold

The sharpest result is the transition between ramp-count = 0 and ramp-count = 5. At zero ramps, 156 blockage events were recorded and every wheelchair user was blocked — 100%. At five ramps, blockages dropped to zero. This complete elimination happened in one step, not gradually. The threshold was not programmed into any agent rule — it emerged from the spatial interaction between agent movement paths and ramp positions along the kerb column across many simulation ticks.

Finding 2 — Accessibility gap persists

Even after blockages are eliminated at ramp-count = 5, wheelchair users consistently take between 68.5 and 76.3 ticks longer than general pedestrians to complete the same corridor. The accessibility gap does not close with ramp provision alone. It reflects the underlying speed difference (wheelchair users: 0.5 m/s vs general pedestrians: 1.2 m/s) combined with the detour cost of navigating to ramp locations.

Finding 3 — Saturation point

From ramp-count = 5 onwards, wheelchair travel time stays between 128.0 and 135.9 ticks regardless of whether 5, 10, 15, or 20 ramps are present — a variation of less than 6%. Adding ramps beyond 5 produces no meaningful improvement in travel time. The bottleneck is no longer ramp availability but agent walking speed and the obstacle layout on the post-kerb section of the corridor.

With random-seed 42 fixing the obstacle layout, this three-zone pattern is a clean result of ramp count alone, not of random variation between runs.

4. Screenshots

Screenshot files are named to reflect the zone and ramp configuration.

Run	Screenshot filename	Zone	What it shows
ramp-count = 0	ramp0_zero_ramps_blockage.png	Exclusion	156 blockages, 100% blocked, red agents stalled at kerb, gap 80.5 ticks
ramp-count = 5	ramp5_transition_zero_blockages.png	Transition	Zero blockages, wheelchair time 132.5 ticks, gap 73.0 ticks

ramp-count = 10	ramp10_saturation_start.png	Saturation	Zero blockages, wheelchair time 135.9 ticks, gap 76.3 ticks
ramp-count = 15	ramp15_saturation_mid.png	Saturation	Zero blockages, wheelchair time 128.0 ticks, gap 68.5 ticks
ramp-count = 20	ramp20_saturation_full.png	Saturation	Zero blockages, wheelchair time 131.8 ticks, gap 72.2 ticks

5. Policy Relevance

- **Zero-ramp streets are fully exclusionary.** 156 blockage events at ramp-count = 0 means most wheelchair users cannot complete the corridor. This directly contravenes the RPwD Act (2016).
- **Five ramps is the minimum viable threshold.** Approximately 15% kerb coverage eliminates all blockages. Concentrated provision above this threshold is more cost-effective than spreading fewer ramps thinly.
- **Ramps alone do not close the mobility gap.** A 68.5 to 76.3 tick disadvantage persists across all ramp configurations. Complementary interventions — obstacle clearance, footpath widening — are needed.
- **Beyond 5 ramps, additional provision has no measurable effect.** Travel time variation across ramp-count 10, 15, and 20 is less than 6%. Infrastructure budgets should prioritise reaching the threshold on more streets.